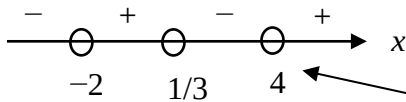
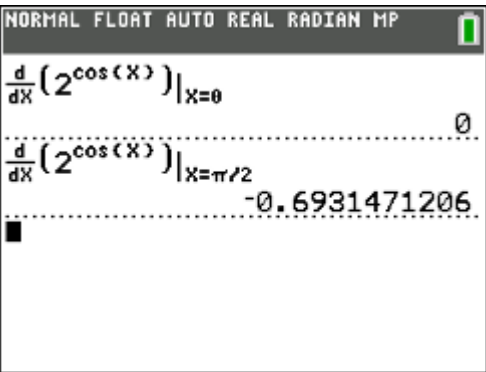


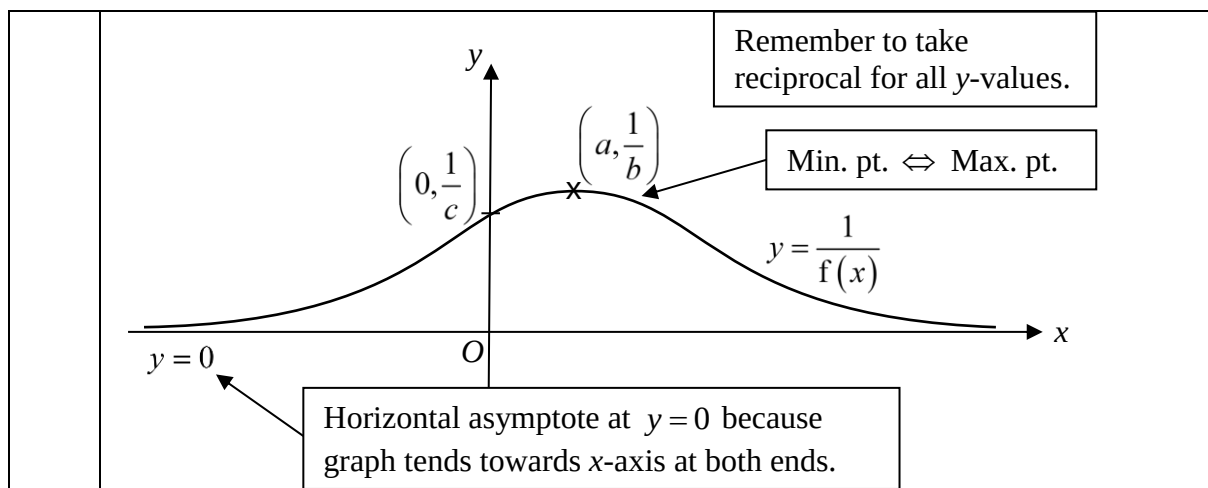
2016 A Level Paper 1 Suggested Solutions

1	Suggested Solutions
	$\frac{4x^2 + 4x - 14}{x - 4} - (x + 3) = \frac{4x^2 + 4x - 14 - (x^2 - x - 12)}{x - 4}$ $= \frac{4x^2 + 4x - 14 - x^2 + x + 12}{x - 4}$ $= \frac{3x^2 + 5x - 2}{x - 4}$ $= \frac{(3x - 1)(x + 2)}{x - 4}$ $\frac{4x^2 + 4x - 14}{x - 4} < x + 3 \quad x \neq 4$ $\frac{4x^2 + 4x - 14}{x - 4} - (x + 3) < 0$ $\frac{(3x - 1)(x + 2)}{x - 4} < 0$  $\therefore x < -2 \text{ or } \frac{1}{3} < x < 4.$ $x = 4$ is the vertical asymptote. Draw number line to solve inequalities. Good habit to label open and closed circles on critical values on number line. Check the sign of each interval.

2	Suggested Solutions
(i)	When $x = 0$, gradient of curve = 0. When $x = \frac{\pi}{2}$, gradient of curve = $-0.69315 = -0.693$ (3 s.f.). 
(ii)	When $x = 0$, gradient of curve = 0 (Tangent is a horizontal line) Equation of tangent: $y = 2^{\cos 0} = 2$ When $x = \frac{\pi}{2}$, $y = 2^{\cos(\frac{\pi}{2})} = 1$ Equation of tangent: Tangent ($y = 2$) is a horizontal line.

$y - 1 = -0.69315 \left(x - \frac{\pi}{2} \right)$ $y = -0.69315x + 2.0888$ (5 s.f.) $y = -0.693x + 2.09$ (3 s.f.) Solving both equations, $2 = -0.69315x + 2.0888$ $x = 0.12810$ Coordinates of point where tangents meet = (0.128, 2) (3 s.f.)	Equation of tangent: $y - y_1 = m(x - x_1)$
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3	Suggested Solutions
	Method 1: $y = f(x) = k(x-l)^4 + m$ Observe that the turning point of $f(x)$ is (l, m). Given that (a, b) is the turning point, $\therefore l = a, m = b$ Sub $(0, c)$ into $y = f(x)$: $k(0-a)^4 + b = c$ $\therefore k = \frac{c-b}{a^4}$ Method 2: $y = f(x) = k(x-l)^4 + m$ $\frac{dy}{dx} = 4k(x-l)^3$ $\frac{dy}{dx} = 0 \Rightarrow x = l$ $\therefore l = a$ Sub (a, b) into $y = f(x)$: $b = k(a-a)^4 + m$ $\therefore m = b$
	From the previous part, $l = a, m = b$ and $k = \frac{c-b}{a^4}$. $y = f(x)$ has y-intercept at $(0, c)$ and turning point at (a, b) Minimum point because $c > b$.



4	Suggested Solutions
(i)	$a + 3d = br^4 \quad \text{----- (1)}$ $a + 8d = br^7 \quad \text{----- (2)}$ $a + 11d = br^{14} \quad \text{----- (3)}$ (2) – (1): $5d = br^4(r^3 - 1)$ (3) – (2): $3d = br^7(r^7 - 1)$ Equating: $\frac{5}{3}br^7(r^7 - 1) = br^4(r^3 - 1)$ $5r^3(r^7 - 1) = 3(r^3 - 1) \quad (\text{since } b, r \neq 0)$ $5r^{10} - 8r^3 + 3 = 0 \quad (\text{shown})$ Using GC, since $r < 1$, $r = 0.74045 \approx 0.74 \quad (2 \text{ d.p.})$ Eliminate a first, then d, followed by b. Always state reason when rejecting answers. Use Polynomial Root Finder (can solve up to degree 10) in PLYSMLT2 App.
(ii)	Required sum $= S_{\infty} - S_n$ $= \frac{b}{1-0.74} - \frac{b}{1-0.74}(1-0.74^n)$ $= \frac{b}{0.26}(1-(1-0.74^n))$ $= \frac{50}{13}b(0.74^n)$ For GP, $S_n = \frac{a}{1-r}(1-r^n) \text{ and}$ $S_{\infty} = \frac{a}{1-r} \text{ if } r < 1$

5	Suggested Solutions
(i)	$\mathbf{u} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} \quad \mathbf{v} = \begin{pmatrix} a \\ 0 \\ b \end{pmatrix}$ $(\mathbf{u} + \mathbf{v}) \times (\mathbf{u} - \mathbf{v}) = \mathbf{u} \times \mathbf{u} - \mathbf{u} \times \mathbf{v} + \mathbf{v} \times \mathbf{u} - \mathbf{v} \times \mathbf{v}$ $= 2(\mathbf{v} \times \mathbf{u})$ Properties of cross product: 1) $\mathbf{u} \times \mathbf{u} = \mathbf{0}$ and $\mathbf{v} \times \mathbf{v} = \mathbf{0}$ because $\mathbf{u} \parallel \mathbf{u}$ and $\mathbf{v} \parallel \mathbf{v}$ 2) $-\mathbf{u} \times \mathbf{v} = \mathbf{v} \times \mathbf{u}$ $= 2 \left(\begin{pmatrix} a \\ 0 \\ b \end{pmatrix} \times \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} \right)$ $= 2 \begin{pmatrix} b \\ 2b - 2a \\ -a \end{pmatrix}$
(ii)	$b = -a$ $(\mathbf{u} + \mathbf{v}) \times (\mathbf{u} - \mathbf{v}) = 2 \begin{pmatrix} -a \\ 2(-a) - 2a \\ -a \end{pmatrix} = -2a \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix}$ $\left -2a \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix} \right = 1$ Unit vector: Vector of magnitude 1. $ -2a \sqrt{18} = 1$ $2a\sqrt{18} = \pm 1$ $a = \pm \frac{1}{6\sqrt{2}}$ Rationalise the denominator. $\therefore a = \frac{\sqrt{2}}{12} \quad \text{or} \quad a = -\frac{\sqrt{2}}{12}$ Hint in question: 'possible values of a', hence there should be more than 1 answer.
(iii)	$(\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v}) = 0$ $\mathbf{u} \cdot \mathbf{u} - \mathbf{u} \cdot \mathbf{v} + \mathbf{v} \cdot \mathbf{u} - \mathbf{v} \cdot \mathbf{v} = 0$ $ \mathbf{u} ^2 - \mathbf{v} ^2 = 0$ Properties of dot product: 1) $\mathbf{u} \cdot \mathbf{u} = \mathbf{u} ^2$ and $\mathbf{v} \cdot \mathbf{v} = \mathbf{v} ^2$ because $\mathbf{u} \parallel \mathbf{u}$ and $\mathbf{v} \parallel \mathbf{v}$ 2) $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$ $ \mathbf{v} = \mathbf{u} \quad \text{since } \mathbf{v} \geq 0$ $ \mathbf{v} = \left \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} \right $ $= \sqrt{2^2 + (-1)^2 + 2^2}$ $= 3$

Qn 6	Suggested Solutions
(ii)	Using $u_0 = 2$ and $u_n = u_{n-1} + n^3 + n$, $u_1 = u_0 + 1^3 + 1 = 2 + 1 + 1 = 4$ $u_2 = u_1 + 2^3 + 2 = 4 + 8 + 2 = 14$ $u_3 = u_2 + 3^3 + 3 = 14 + 27 + 3 = 44$
(iii)	$\begin{aligned} \sum_{r=1}^n (u_r - u_{r-1}) &= \cancel{u_1} - u_0 \\ &\quad + \cancel{u_2} - \cancel{u_1} \\ &\quad + \cancel{u_3} - \cancel{u_2} \\ &\quad + \dots \\ &\quad + \cancel{u_n} - \cancel{u_{n-1}} \\ &= u_n - u_0 \end{aligned}$ Note: from (ii), $u_n = u_{n-1} + n^3 + n$ $u_n - u_{n-1} = n^3 + n$ $\begin{aligned} \sum_{r=1}^n (u_r - u_{r-1}) &= \sum_{r=1}^n (r^3 + r) \\ u_n - u_0 &= \sum_{r=1}^n r(r^2 + 1) \\ u_n - 2 &= \frac{1}{4} n(n+1)(n^2 + n + 2) \quad (\text{from part (i)}) \\ u_n &= \frac{1}{4} n(n+1)(n^2 + n + 2) + 2 \end{aligned}$

7	Suggested Solutions
(a)	Sub $w = -1 + 5i$ into LHS $(-1 + 5i)^2 + (-1 - 8i)(-1 + 5i) + (-17 + 7i)$ $= 1 - 10i - 25 + 1 - 5i + 8i + 40 - 17 + 7i$ $= 0$ Therefore, $-1 + 5i$ is a root of the equation. Let z be the second root. $w^2 + (-1 - 8i)w + (-17 + 7i) = (w - (-1 + 5i))(w - z)$ By comparing the coefficient of w : $-1 - 8i = -(-1 + 5i) - z$ $z = -(-1 + 5i) + 1 + 8i$ $= 2 + 3i$ Therefore, the second root is $2 + 3i$.
(b)	Since coefficients of polynomial are all real, $z = 1 + ai$ is a root $\Rightarrow z = 1 - ai$ is also a root.

Verify that LHS = RHS.

Must state reason why conjugate pair is also a root.

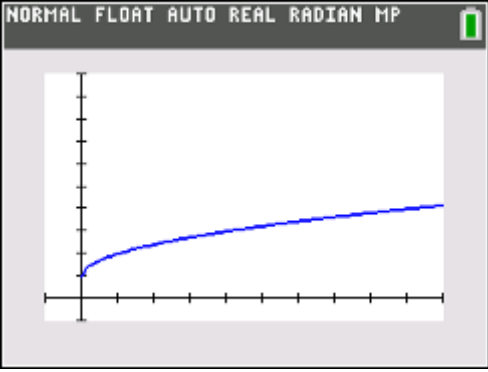
Since the polynomial is of degree 3, the last root must be purely real. Let $z = b$ be the third root. $z^3 - 5z^2 + 16z + k = (z - (1 + ai))(z - (1 - ai))(z - b)$ $= ((z - 1) - ai)((z - 1) + ai)(z - b)$ $= ((z - 1)^2 - (ai)^2)(z - b) \leftarrow \begin{array}{l} \text{Using the algebraic property} \\ (A - B)(A + B) = A^2 - B^2 \end{array}$ $= (z^2 - 2z + 1 + a^2)(z - b)$ Comparing coefficient of z^2: $-b - 2 = -5 \Rightarrow b = 3$ Comparing coefficient of z: $-2(-3) + (1 + a^2) = 16 \Rightarrow a = 3 \quad (\because a > 0)$ Comparing constant: $k = (1 + 3^2)(-3) \Rightarrow k = -30$ $\therefore a = 3$ and $k = -30$.

8	Suggested Solutions
(i)	$f(x) = \tan(ax + b)$ $f'(x) = a \sec^2(ax + b)$ $= a(1 + \tan^2(ax + b)) \leftarrow \begin{array}{l} \text{Using the trigonometry identity} \\ \sec^2 \theta = 1 + \tan^2 \theta \end{array}$ $= a(1 + y^2)$ $= a + ay^2 \quad (\text{shown})$ $f''(x) = 2ayf'(x) \leftarrow \begin{array}{l} \text{Implicit differentiation} \end{array}$ $= 2ay[a + ay^2]$ $= 2a^2y(1 + y^2)$ $f'''(x) = 2a((f'(x))^2 + yf''(x)) \leftarrow \begin{array}{l} \text{Product rule} \end{array}$ $= 2a((a(1 + y^2))^2 + y(2a^2y(1 + y^2)))$ $= 2a^3(1 + y^2)[(1 + y^2) + 2y^2]$ $= 2a^3(1 + y^2)(1 + 3y^2)$
(ii)	When $b = \frac{\pi}{4}$, $f(0) = \tan\left(\frac{\pi}{4}\right) = 1 = y$ $f'(0) = a + a(1)^2 = 2a$ $f''(0) = 2a^2(1)(1 + 1^2) = 4a^2$ $f'''(0) = 2a^3(1 + 1^2)(1 + 3(1)^2) = 16a^3$ $f(x) = 1 + 2ax + \frac{4a^2}{2}x^2 + \frac{16a^3}{3!}x^3 + \dots$ $= 1 + 2ax + 2a^2x^2 + \frac{8a^3}{3}x^3 + \dots \quad (\text{up to } x^3)$ Maclaurin expansion: $f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots + \frac{x^n}{n!}f^{(n)}(0) + \dots$

(iii)	When $a = 2, b = 0, f(x) = \tan(2x + 0) = \tan 2x$ $f(0) = \tan(0) = 0 = y$ $f'(0) = 2 + 2(0)^2 = 2$ $f''(0) = 2(2)^2(0)(1 + 0^2) = 0$ $f'''(0) = 2(2)^3(1 + 0^2)(1 + 3(0)^2) = 16$ $\tan 2x = 2x + \frac{16}{3!}x^3 + \dots$ $= 2x + \frac{8}{3}x^3 + \dots$ (first 2 non-zero terms) Maclaurin expansion: $f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \dots + \frac{x^n}{n!} f^{(n)}(0) + \dots$
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9	Suggested Solutions
(i)(a)	$y = \frac{dx}{dt} \quad \text{--- (1)}$ $\frac{dy}{dt} = \frac{d^2x}{dt^2} \quad \text{--- (2)}$ Sub (1) and (2) into given D.E. $\frac{d^2x}{dt^2} + 2\frac{dx}{dt} = 10$ $\frac{dy}{dt} + 2y = 10$ $\frac{dy}{dt} = 10 - 2y$ (shown) Show every step clearly.
(i)(b)	$\frac{dy}{dt} = 10 - 2y$ $\int \frac{1}{y-5} dy = \int -2 dt \quad \leftarrow \text{Variable separable}$ $\ln y-5 = -2t + c, c \in \mathbb{R}$ $y-5 = e^{-2t+c}$ $y-5 = \pm e^{-2t+c} \quad \leftarrow \text{Remember to get rid of the } \pm \text{ sign.}$ $y = 5 + Ae^{-2t}, A = \pm e^c$ Given $x = 0$ and $\frac{dx}{dt} = 0$ when $t = 0$: $\Rightarrow t = 0, y = 0 \quad \leftarrow y = \frac{dx}{dt}$ $0 = 5 + Ae^{-2(0)}$ $A = -5$ $\therefore y = 5 - 5e^{-2t}$ Given that $y = \frac{dx}{dt}$: $\frac{dx}{dt} = 5 - 5e^{-2t}$ $x = 5t + \frac{5}{2}e^{-2t} + d, d \in \mathbb{R}$

	Given $x = 0$ and $\frac{dx}{dt} = 0$ when $t = 0$: $0 = 5(0) + \frac{5}{2}e^{-2(0)} + d$ $d = -\frac{5}{2}$ $\therefore x = 5t + \frac{5}{2}e^{-2t} - \frac{5}{2}$
(ii)	$\frac{d^2x}{dt^2} = 10 - 5\sin\frac{1}{2}t$ $\frac{dx}{dt} = 10t + 10\cos\frac{1}{2}t + g \quad \text{----(3)}$ $x = 5t^2 + 20\sin\frac{1}{2}t + gt + f \quad \text{----(4)}$ Given $x = 0$ and $\frac{dx}{dt} = 0$ when $t = 0$: From (3): $g = -10$ From (4): $f = 0$ $\therefore x = 5t^2 + 20\sin\frac{1}{2}t - 10t$
(iii)	Using GC, when $x = 5$, For Model (i): $5 = 5t + \frac{5}{2}e^{-2t} - \frac{5}{2}$ ← Time taken = $1.47377 \approx 1.47\text{s}$ (2 d.p.) For Model (ii): $5 = 5t^2 + 20\sin\frac{1}{2}t - 10t$ Time taken = $1.04597 \approx 1.05\text{s}$ (2 d.p.) Sketch graphs $Y_1 = 5$ and $Y_2 = 5t + \frac{5}{2}e^{-2t} - \frac{5}{2}$, then find the intersection point.

10	Suggested Solutions
(a)(i)	$y = f(x)$ $y = 1 + \sqrt{x}$  $x = (y - 1)^2$ $f^{-1}(x) = (x - 1)^2$ $D_{f^{-1}} = R_f = [1, \infty)$ $f^{-1}: x \mapsto (x - 1)^2, x \geq 1$
(a)(ii)	$ff(x) = x$ ----- (1) $1 + \sqrt{1 + \sqrt{x}} = x$ (note that $x > 2$ because $1 + \sqrt{1 + \sqrt{x}} > 2$) $1 + \sqrt{x} = (x - 1)^2$ $x = (x^2 - 2x + 1 - 1)^2$ $1 = x(x - 2)^2$ (since $x > 2$) $x^3 - 4x^2 + 4x - 1 = 0$ (shown) From GC, $x = 2.62$ or $x = 1$ or $x = 0.382$ (3 s.f.) (rej $\because x > 2$) (rej $\because x > 2$) Alternative method to reject is to sub the values found into equation (1), and show that they do not satisfy equation (1). Method 1: Since $R_{ff} \subseteq R_f = D_{f^{-1}}$, $f^{-1}ff$ exists. $ff(2.62) = 2.62$ $f^{-1}ff(2.62) = f^{-1}(2.62)$ $f(2.62) = f^{-1}(2.62)$ $\therefore x = 2.62$ satisfies $f(x) = f^{-1}(x)$. Method 2: $ff(x) = x$ implies that $f(x) = f^{-1}(x)$ on $D_{f^{-1}} \cap D_f$. Since $2.62 \in D_{f^{-1}} \cap D_f$, $x = 2.62$ satisfies $f(x) = f^{-1}(x)$.

(b)(i)	$g(0) = 1$ $g(1) = 1 + g(0) = 1 + 1 = 2$ $g(2) = 2 + g(1) = 2 + 2 = 4$ $g(3) = 1 + g(2) = 1 + 4 = 5$ $g(4) = 2 + g(3) = 2 + 5 = 7$ $g(5) = 1 + g(4) = 1 + 6 = 7$ $g(6) = 2 + g(3) = 2 + 5 = 7$ $g(7) = 1 + g(6) = 1 + 7 = 8$ $g(12) = 2 + g(6) = 2 + 7 = 9$
(b)(ii)	$\therefore g(5) = g(6) = 7$ from (bi) Therefore g is not 1-1, implying that g does not have an inverse. To prove that a function is not 1-1, we need to give counter-example. I.e. Find 2 different values of x that give the same y-value.

11	Suggested Solutions
(i)(a)	Method 1: $\text{Normal vector} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \times \begin{pmatrix} a \\ 4 \\ -2 \end{pmatrix}$ $= \begin{pmatrix} -4 \\ 2 \\ 4 - 2a \end{pmatrix}$ When $a = 0$, $\therefore \begin{pmatrix} -4 \\ 2 \\ 4 \end{pmatrix} = 2 \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}$ $\therefore$ normal vector of plane is parallel to direction vector of line $\Rightarrow l$ is perpendicular to p. Solving equations of l and p, $\begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 4 \\ -2 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + t \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}$ $\lambda + 2t = -2$ $2\lambda + 4\mu - t = 3$ $-2\mu - 2t = -1$ Using GC, $\lambda = -\frac{8}{9}, \mu = \frac{19}{18} \text{ and } t = -\frac{5}{9}$ Method 2: when $a = 0$, $\begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} \cdot \left(\lambda \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 4 \\ -2 \end{pmatrix} \right)$ $= -2\lambda + 2\lambda + 4\mu - 4\mu$ $= 0$ Therefore, l is perpendicular to p.

	Position vector of point of intersection = $\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} - \frac{5}{9} \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = \frac{1}{9} \begin{pmatrix} 1 \\ -5 \\ -1 \end{pmatrix}$ Coordinates of point of intersection: $\left(\frac{1}{9}, -\frac{5}{9}, -\frac{1}{9}\right)$
(i)(b)	$\mathbf{r} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}$ $= -2 - 3 + 4 = -1$ $p : \mathbf{r} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = -1$ Method 1: Let $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ be a point on a plane 12 units from p. $\frac{\left \left(\begin{pmatrix} x \\ y \\ z \end{pmatrix} - \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} \right) \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} \right }{\sqrt{(-2)^2 + 1^2 + 2^2}} = 12$ $\begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} - (-1) = \pm 36$ $-2x + y + 2z = -1 \pm 36$ The Cartesian equations of the planes are $-2x + y + 2z = 35$ and $-2x + y + 2z = -37$. Method 2: Standard form: $\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}$ $p : \mathbf{r} \cdot \frac{1}{3} \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = \frac{-1}{3}$ Note that $\mathbf{a} \cdot \mathbf{n}$ is the perpendicular distance from the origin, O, to the plane. Thus the two planes are: $\mathbf{r} \cdot \frac{1}{3} \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = \frac{-1}{3} \pm 12$ The Cartesian equations of the planes are $-2x + y + 2z = 35$ and $-2x + y + 2z = -37$.
(ii)	From (i), Normal vector = $\begin{pmatrix} -4 \\ 2 \\ 4 - 2a \end{pmatrix}$ For l and p to not meet in a unique point, l must be parallel to p. $\mathbf{n} \cdot \mathbf{d} = 0$ $\begin{pmatrix} -4 \\ 2 \\ 4 - 2a \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = 0$ $8 + 2 + 8 - 4a = 0$ $a = \frac{9}{2}$